Process & Decision Documentation

Process Overview Visualization (Group Work Only)

The "Octopus Iteration Loop"

- Early Phase (Research & Assets): Visual Artist creates the café background and octopus sprites. Game Designer defines the 5 mini-games (Hold, Tap, Balance, etc.).
- Mid Phase (Core Integration): Integrates assets into p5.js.
    - *Pivot Point:* Realized the UI was too small on high-res screens.
    - *Action:* Massive UI scaling and "Current Order" card redesign.
- Late Phase (Mechanical Depth): Implementation of the Stress System.
    - *Feedback Loop:* Playtesting showed that "Tangles" were too easy to ignore.
    - *Decision:* Added "Physical Tentacle Crossing" logic and the "Hold R to Calm" mechanic.
- Final Phase (Onboarding):
    - *Pivot Point:* Instructions were too confusing mid-game.
    - *Action:* Added a post-transition Training Manual screen.

Project/Assignment Decisions

Major Decision	**What Changed**	**Why?**	**Who Involved**
UI Scaling & Layout	Rescaled all panels (1.5x) and swapped the Syrup and Servestations.	The original UI was unreadable. Swapping stations prevented the Serve panel from overlapping with the Shift Info timer.	Entire Group
Stress-Tangle Coupling	Tied the Tangle meter to the Stress level. Tangle now increases automatically when Stress is >40.	To create a "chaos loop." It simulates the feeling of being overwhelmed (Dyspraxia simulation) rather than just a simple timer.	Entire Group

The "Training Manual"	Added a static instruction screen after the loading animation but before the first order.	Initial play testers tried to drag tentacles without knowing which keys to press. We needed a "safe space" for players to read rules.	Entire Group
Physical Crossing	Added CheckTentacleCrossingusing line-segment intersection math.	Originally, "Tangles" were just a random stat. Adding physical crossing made player movement and "cable management" a core skill.	Entire Group
Continuous Challenge	Revise the little challenge games from one after another completion to on going challenge until the drink is done.	The original mechanics where player will only continue to the next challenge after one is complete is too easy after player figure out the game mechanics for each challenge, which can't fully demonstrate the struggle of dyspraxia because it's still too organized, therefore after reflecting feedback from professor/student playtesting, we decide to change the whole game mechanics to a ongoing challenges layer up.	Entire Group
Tentacle Interaction	We changed the tentacle controlling mechanics from using WASD to mouse dragging	According to student and professor feedback, WASD controlling create a disconnected affordance, because tentacles are visually vertically aligned, therefore it's very likely for them to misinterpreted that	Entire Group

		they need to use WASD to control the tentacles instead of select one. Hence, we change from using keys to select tentacle to use mouse to select and drag, which not only make the affordance more connected, but also better showcased Dyspryxia stress.	
Difficulty Progression	Introduced progressive recipe complexity (1 → 4 steps).	From the playtesting, most player struggle to successfully make the coffee that require passing multiple challenges. We realized that playtesters needed time to learn mechanics before facing multitasking overload. Therefore, we added a level system where difficulties will be categorized based on steps required to make coffe. (e.g. 1 steps are all level 1, 2 steps are all level 2, and so on)	Entire Group
Level Order Debugging	Fixed a bug where Level 1 would repeat twice before progressing to Level 2.	The repeated Level 1 disrupted the intended game progression and confused players during onboarding. It created inconsistency in difficulty scaling and made the experience feel boring. Fixing this ensured a smooth, predictable	Entire Group

		progression from Level to Level.	
Score System Debugging	Redesigned the scoring system to be rule-based and level-scaled: per-order points increase from +10 (Level 1) to +40 (Level 4), full level completion awards +30–50 points, and each tangling mistake deducts 5 points.	The original scoring system was chaotic and added points inconsistently, making it difficult for players to understand how their performance translated into scores. This reduced motivation and clarity. Standardizing the scoring rules created a clear feedback loop, improved fairness, and helped players better track progress and success.	Entire Group

A1 – A3 (Group Work)

Provide more details, as decisions are shared, cumulative, and affect multiple roles.

Document decisions such as:

- Scope changes or feature cuts
- Design or analytical pivots
- Outcomes of playtesting or peer/instructor feedback
- Technical or coordination-related trade-offs
- Ethical, representational, or global contexts for design decisions

For each major decision, briefly include:

- What changed
- Why the decision was made
- Who was involved or responsible

This may be written in short paragraphs or bullet points.

This section is not a complete project history. It is a record of decisions that shape the work. Clarity, judgement, and relevance matter more than length.

Role-Based Process Evidence

This section documents how your work developed over time and provides concrete evidence of your process. It is used to make your thinking, iteration, and decision-making visible, whether or not GenAI was used. You may reuse relevant process evidence across documents where appropriate, as long as it accurately reflects the work for that assignment.

For group assignments (A1 – A3), this section is completed collectively, with clearly attributed role-based entries. For the group, your entries should correspond to elements shown in your process overview visualization such as stages, iterations, or decision points.

For individual assignments (Side Quests and A4), this section is completed individually, using a simplified version of the same structure. For Side Quests, you may omit sections below where no meaningful decision-making occurred. For example, for the first side quest which you need to just upload a blank p5.js to GitHub, you must have to remove a significant amount of the process based evidence as not a lot of design decisions were made.

Examples of acceptable process evidence include (but are not limited to):

- Screenshots of video editing timelines
- Drafts of scripts or research notes
- Annotated sources
- Sketches or diagrams
- Early document outlines
- Prototype screenshots
- Code snippets or commits
- Before-and-after revisions

Project management artifacts (e.g., task boards, timelines, coordination) may be included here when they are relevant to understand how assignment-level decisions were made.

If GenAI was used to support planning, coordination, or role management, this should be noted briefly here as part of the process evidence. Deeper reflection on how GenAI affected teamwork, fairness, and coordination over time belongs in the GenAI Reflections (A4).

Entry Header

Starter

Name: Nicole Wu

Role(s): Initial Game Developer & Visual Designer

Primary responsibility for this work: Building the Starter Version of the game, establishing the core p5.js environment, and implementing the first iteration of basic station tasks and recipe logic. Additionally, developed the foundational visual elements, including game characters, background art, and basic UI/UX design.

Name: Suyao Liu

Role(s): Initial Game Developer & Asset Manager

Primary responsibility for this work: Developing the foundational logic for the **Starter Version** and managing the initial integration of visual assets, background placement, and basic station templates.

Research & Feedback

Name: Sofia Xu

Role(s): Research & User Experience Analyst

Primary responsibility for this work: Leading the **Research Phase**, conducting playtesting sessions to identify user pain points, and synthesizing feedback to guide mechanical balancing.

Name: Casey Shen

Role(s): Feedback Coordinator & Quality Assurance

Primary responsibility for this work: Analyzing feedback from playtests and peer reviews to identify **Pivot Points**, ensuring the game's difficulty curve met project goals, and documenting user needs.

Refinement

Name: Emily He

Role(s): Technical Refinement & UI/UX Developer

Primary responsibility for this work: Developing the **Refined Version** of the game, including implementing the Stress/Tangle systems, complex math for tentacle crossing, and universal UI scaling for accessibility.

Name: Fiona Luo

Role(s): Technical Refinement & System Integration

Primary responsibility for this work: Collaborating on the **Refined Version** to ensure code stability, optimizing the mini-game mechanics (Balance/Rhythm), and refining the state-machine transitions between the manual and gameplay.

Goal of Work Session

The primary goal was to develop the **Minimum Viable Product (MVP)** of the game. We focused on establishing the core p5.js environment, importing the initial café assets, and coding the basic logic for the first three stations (Coffee, Milk, and Ice) to ensure the fundamental "drag-and-drop" mechanic functioned correctly before handing it off for refinement.

Tools, Resources, or Inputs Used

- **Prior drafts or code:** Initial p5.js boilerplate.
- **Lecture Notes:** Referenced course material on basic state machines and event listeners.
- **Teammates:** Constant communication with the artists to ensure asset dimensions matched the canvas.
- **External references:** P5.js Reference Documentation for coordinate mapping.

GenAI Documentation

If GenAI was used (keep each response as brief as possible):

Date Used: February 26, 2026 – March 12, 2026

Tool Disclosure: Gemini 3 Pro, Microsoft Copilot, ChatGPT 5.2

Purpose of Use: Debugging p5.js logic, calculating vector math for tentacle crossing detection, and refactoring global UI scaling code.

Summary of Interaction: The tool provided the algorithm for line-segment intersection (doSegmentsIntersect) and assisted in converting static UI coordinates into dynamic coordinates based on the background scale.

Human Decision Point(s): The AI initially suggested making the "Tangle" mechanic a random probability. We overruled this and insisted on a physics-based crossing detection. This decision was made to ensure the game served as a true "coordination simulation," where player actions have direct, deterministic consequences.

Integrity & Verification Note: We verified the output by conducting manual "stress tests" within the game, intentionally crossing tentacles to see if the state triggered correctly and resizing the browser window to ensure the UI remained responsive.

Scope of GenAI Use: GenAI did not contribute to the creation of the initial game concept, or the creative narrative/quotes.

Limitations or Misfires: The AI initially failed to account for the hardware "Caps Lock" state on different operating systems, which caused the Ice station logic to fail. We manually added TAB as a secondary input to fix this.

Summary of Process (Human + Tool)

Based on the most recent playtesting feedback, we identified critical issues in both level progression and the scoring system, and implemented targeted improvements to enhance gameplay clarity and consistency.

1. **Level Order Debugging:**
   Fiona and Emily focused on resolving a progression bug where Level 1 repeated twice before advancing. They removed the unintended level, ensuring a smooth and logical level sequence that better supports player onboarding and difficulty scaling.

2. **Score System Redesign:**
   Selina and Nicole redesigned the scoring system to address inconsistencies observed during playtesting. The updated system introduces clear, level-scaled and rule-based scoring: per-order points increase progressively from +10 (Level 1) to +40 (Level 4), while full level completion awards +30–50 points depending on the level. Additionally, each tangling mistake now deducts 5 points to incorporate performance-based penalties. This creates a more transparent, fair, and motivating feedback loop for players.

3. **Documentation & Synthesis:**
   Casey and Sofia consolidated all post-playtesting insights, summarized key changes, and updated the written documentation to accurately reflect the latest iteration of the project.

These updates ensure the game experience is more intuitive, consistent, and aligned with player expectations, while also improving internal clarity for future development.

Decision Points & Trade-offs

Level Order Fix:

Issue: Level 1 unintentionally repeated twice before progressing to Level 2.

Action: Removed the duplicate level trigger to ensure correct level sequencing.

Reasoning: This was a system bug rather than a design choice. Fixing it restored a smooth and predictable progression, improving onboarding and maintaining the intended difficulty curve.

Score System Redesign:

Issue: The original scoring system was inconsistent and unclear, with points being added unpredictably.

Action: Implemented a structured, level-scaled scoring system:

- Level 1: +10 per order, +30 per level
- Level 2: +20 per order, +30 per level
- Level 3: +30 per order, +50 per level
- Level 4: +40 per order, +50 per level
- −5 points per tangling mistake

Reasoning: This was not a trade-off between alternatives, but a necessary redesign to establish clarity, fairness, and a meaningful feedback loop for player performance.

Verification & Judgement

- **Playtesting:**
  We observed that players felt more in control after introducing the leveling system along with a tutorial level that excluded stress and tangle mechanics. This staged

onboarding helped players build confidence and familiarity before entering more complex gameplay. As a result, players were more calm and capable when handling orders in real levels, and showed increased motivation to continue challenging themselves once they understood their performance. However, playtesters also expressed confusion about the scoring system, as they did not clearly understand how points were calculated or how they could improve their score.

- **Peer Feedback:**
  Based on group discussion and peer feedback, we identified key issues in both the scoring system and level progression. As a result, we redesigned the scoring system to make it more consistent and understandable, and fixed a bug causing incorrect level sequencing. These changes directly addressed player confusion and improved overall gameplay clarity.

- **Accessibility Check:**
  We evaluated the game against universal design principles. By adding the TAB key as a backup for Caps Lock, we ensured that users with different keyboard hardware could still complete the Ice challenge.

Limitations, Dead Ends, or Open Questions

- **Dead Ends:**
  We attempted to implement 3D-shaded tentacles using p5.js WebGL mode. However, this caused a major performance drop and conflicted with our 2D station icons. We retreated to 2D Bezier curves to prioritize smooth gameplay.

- **Unresolved Questions:**
  We discussed whether the background music's pitch should increase as Stress rises. Due to time constraints, this remained unresolved, and we focused on visual "shaking" instead.

- **Compromise:**
  To prevent the Tangle mechanic from being too frustrating, we used a slightly wider collision box than the actual tentacle lines. This was a deliberate choice to balance realism with player "fun."

- **UI Limitation:**
  The initial UI layout generated with AI tools was overly dense and ineffective. It

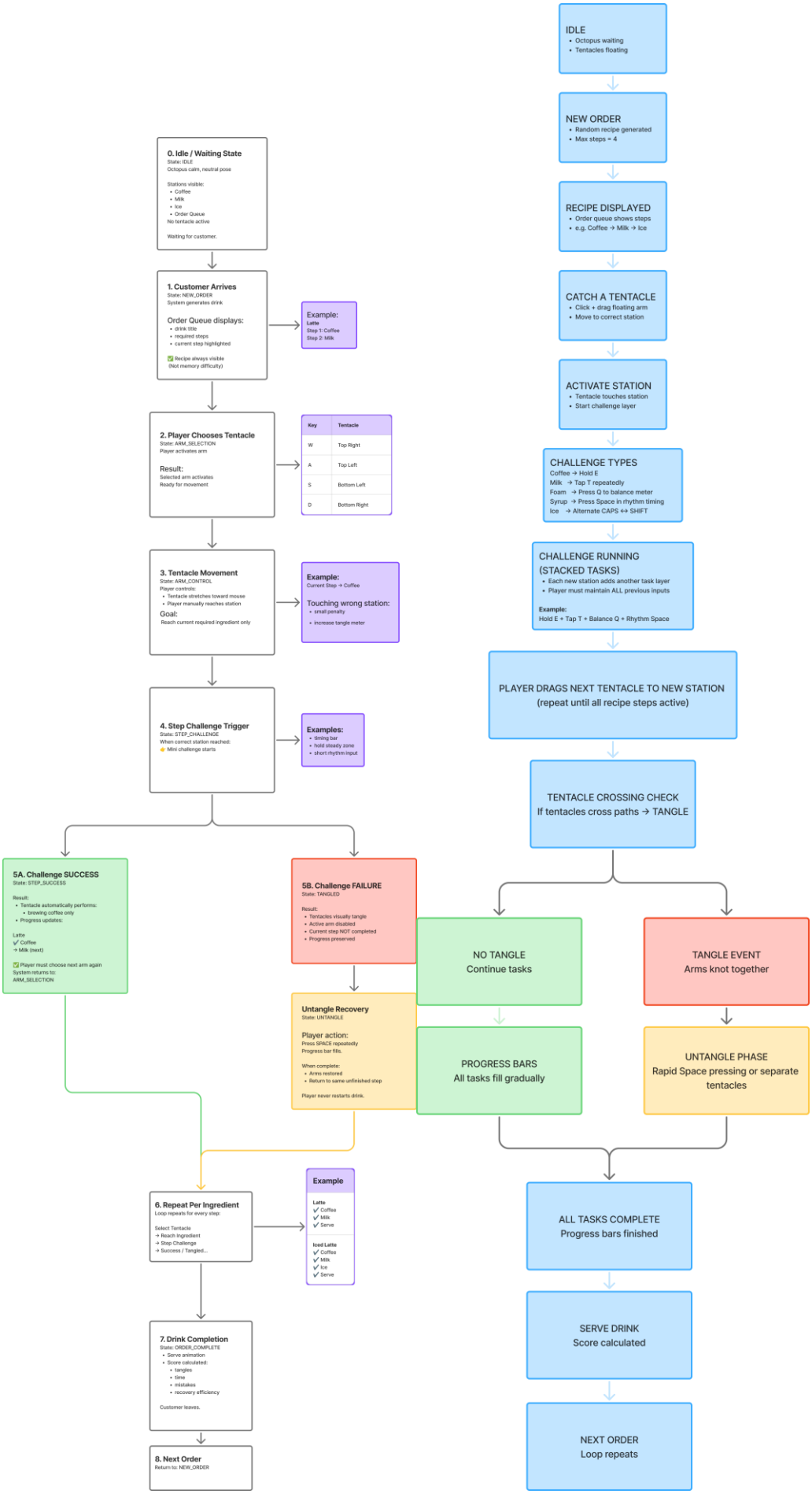
attempted to present too much information at once, likely assuming that more explanation would improve player understanding. In practice, this overwhelmed players and made it difficult to identify where to focus. This highlighted the limitation of AI-generated interfaces without human-centered refinement, and the need to prioritize visual hierarchy and clarity over information quantity.

Appendix

https://gemini.google.com/share/64e62ff5f65f

https://chatgpt.com/share/69b38ffc-6024-8005-bd02-4fedf27dd3da

https://chatgpt.com/share/69dc5b95-1e10-832b-b31f-12fa24a6d165


0. Idle / Waiting State
State: IDLE
Octopus calm, neutral pose
Stations visible:
• Coffee
• Milk
• Ice
• Order Queue
No tentacle active
Waiting for customer.
1. Customer Arrives
State: NEW_ORDER
System generates drink
Order Queue displays:
• drink title
• required steps
• current step highlighted
✅ Recipe always visible
(Not memory difficulty)
Example:
Latte
Step 1: Coffee
Step 2: Milk
2. Player Chooses Tentacle
State: ARM_SELECTION
Player activates arm
Result:
Selected arm activates
Ready for movement
Key | Tentacle
W | Top Right
A | Top Left
S | Bottom Left
D | Bottom Right
3. Tentacle Movement
State: ARM_CONTROL
Player controls:
• Tentacle stretches toward mouse
• Player manually reaches station
Goal:
Reach current required ingredient only
Example:
Current Step → Coffee
Touching wrong station:
• small penalty
• increase tangle meter
4. Step Challenge Trigger
State: STEP_CHALLENGE
When correct station reached:
👉 Mini challenge starts
Examples:
• timing bar
• hold steady zone
• short rhythm input
5A. Challenge SUCCESS
State: STEP_SUCCESS
Result:
• Tentacle automatically performs:
• brewing coffee only
• Progress updates:
Latte
✓ Coffee
→ Milk (next)
✅ Player must choose next arm again
System returns to:
ARM_SELECTION
5B. Challenge FAILURE
State: TANGLED
Result:
• Tentacles visually tangle
• Active arm disabled
• Current step NOT completed
• Progress preserved
Untangle Recovery
State: UNTANGLE
Player action:
Press SPACE repeatedly
Progress bar fills.
When complete:
• Arms restored
• Return to same unfinished step
Player never restarts drink.
6. Repeat Per Ingredient
Loop repeats for every step:
Select Tentacle
→ Reach Ingredient
→ Step Challenge
→ Success / Tangled...
Example
Latte
✓ Coffee
✓ Milk
✓ Serve
Iced Latte
✓ Coffee
✓ Milk
✓ Ice
✓ Serve
7. Drink Completion
State: ORDER_COMPLETE
• Serve animation
• Score calculated:
• tangles
• time
• mistakes
• recovery efficiency
Customer leaves.
8. Next Order
Return to: NEW_ORDER
IDLE
• Octopus waiting
• Tentacles floating
NEW ORDER
• Random recipe generated
• Max steps = 4
RECIPE DISPLAYED
• Order queue shows steps
• e.g. Coffee → Milk → Ice
CATCH A TENTACLE
• Click + drag floating arm
• Move to correct station
ACTIVATE STATION
• Tentacle touches station
• Start challenge layer
CHALLENGE TYPES
Coffee → Hold E
Milk → Tap T repeatedly
Foam → Press Q to balance meter
Syrup → Press Space in rhythm timing
Ice → Alternate CAPS ↔ SHIFT
CHALLENGE RUNNING (STACKED TASKS)
• Each new station adds another task layer
• Player must maintain ALL previous inputs
Example:
Hold E + Tap T + Balance Q + Rhythm Space
PLAYER DRAGS NEXT TENTACLE TO NEW STATION
(repeat until all recipe steps active)
TENTACLE CROSSING CHECK
If tentacles cross paths → TANGLE
NO TANGLE
Continue tasks
TANGLE EVENT
Arms knot together
PROGRESS BARS
All tasks fill gradually
UNTANGLE PHASE
Rapid Space pressing or separate tentacles
ALL TASKS COMPLETE
Progress bars finished
SERVE DRINK
Score calculated
NEXT ORDER
Loop repeats

Stage Progression Rules

Difficulty increases based on successful order completions.
Example progression:
Stage 1 → 3 successful orders → unlock Stage 2
Stage 2 → 3 successful orders → unlock Stage 3
Stage 3 → 3 successful orders → unlock Stage 4
This ensures players feel gradual mastery before difficulty increases.

Visual Feedback

When a new stage unlocks, briefly display:
New Recipe Complexity Unlocked!
This gives players a clear sense of progression.

Stage 1: Tutorial Stage (1 Step Recipes)

Only single-step drinks appear.

Purpose:
- teach tentacle dragging
- teach how each station challenge works
- allow players to focus on one input mechanic

Player only manages one active challenge at a time.

- Americano
  1. Coffee
- Milk Sample
  1. Milk
- Foam Practice
  1. Foam
- Ice Practice
  1. Ice

Stage 2: Beginner Multitasking (2 Step Recipes)

After several successful drinks, recipes expand to two steps.

Player now experiences two simultaneous challenges.
Example situation:
- Hold E
- Tap T

This introduces the concept of layered coordination.

- Latte
  1. Coffee
  2. Milk
- Iced Coffee
  1. Coffee
  2. Ice

Stage 3: Intermediate Difficulty (3 Step Recipes)

Once the player demonstrates consistent success, recipes increase to three steps.

Players must manage three ongoing motor tasks simultaneously.
Example:
- Hold E
- Tap T
- Press Q

Tentacle tangling becomes more likely due to increased movement.

- Iced Latte
  1. Coffee
  2. Milk
  3. Ice
- Foam Latte
  1. Coffee
  2. Milk
  3. Foam

Stage 4: Full Challenge (4 Step Recipes)

The highest difficulty stage includes recipes with four ingredients.

Players now manage four simultaneous challenges, which creates the intended chaotic experience.
Example situation:
- Hold E
- Tap T
- Balance Q
- Alternate Caps ↔ Shift

This stage fully expresses the dyspraxia coordination overload the game is designed to simulate.

- Special Latte
  1. Coffee
  2. Milk
  3. Foam
  4. Syrup
- Iced Deluxe Latte
  1. Coffee
  2. Milk
  3. Foam
  4. Ice